

Quantum Information Theory

Lesson 7: Quantum Compression

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Remember: Classical Compression

The key insight when we discussed classical compression was to look at any input message $\mathbf{w}$ as a stream of i.i.d. samples of a random variable X called a memoryless source. This X produced strings of message tokens $m_0, \dots, m_n \in \mathcal{M}$, so that $\mathbf{w}$ would be one of the possible message strings obtained from X .

To see how much that information could be compressed, we considered a variable-length encoder $f : \mathcal{M} \rightarrow \mathbb{Z}_d^*$ such as the Huffman algorithm, and we wanted to shrink the expected codeword length while still being able to invert (decode) f .

Shannon's source coding theorem looked at what happened when we tried to stream possibly infinitely long streams of samples of X . It stated that the optimal average codeword length satisfies

$$\lim_{n \rightarrow \infty} \frac{1}{n} |\mathbb{E}[L] - nH(X)| = 0,$$

a good asymptotic approximation of the true optimum, that for finite n is numerically computed analyzing the Huffman algorithm.

Remember: Classical Compression

To prove Shannon's coding theorem, we looked at what happens if we try to use less dits per message token, and we found that if we used less then there is no decoder that we can use to invert f .

To handle the happy path, we noticed that in a string of n tokens, the frequency of each possible token n_i is clustered around $np_i \pm \epsilon$ so to give our encoder some leeway we were even open to let our decoder fail when the input was not as just described, ϵ -typical. If we let our decoder fail with probability π then we could ignore ϵ -atypical inputs with $\epsilon = 1/\sqrt{\pi n}$. But, alas, this was not enough.

Indeed, as long as $\pi \neq 1$, we would need $\mathbb{E}[L] \geq nH(X) + o(n)$. More precisely, we found that if we are open to let our encoder use on average $H(X) + \delta$ dits per token, then there is an $\epsilon \leq \delta/C_{X,d}$ and a minimal packed amount $n \geq 1/(\pi\epsilon^2)$ of letters for which such an encoder can be found to fail with probability less than π .

Quantum State Discrimination

Before creating a procedure to compress quantum states, we would first need to be able to tell if what we decoded was any good.

For a toy setting, let's say what we decoded is, with probability p , the ρ that we encoded, and, with probability $1 - p$, some fixed σ .

To distinguish them, we use a POVM with operators $\mathbf{F}$ and $\mathbf{I} - \mathbf{F}$. The probability to correctly identify what was sent is

$$\begin{aligned} p \operatorname{Tr}(\mathbf{F}\rho) + (1 - p) \operatorname{Tr}((\mathbf{I} - \mathbf{F})\sigma) &= 1 - p + \operatorname{Tr}(\mathbf{F}(p\rho - (1 - p)\sigma)) \\ &= p - \operatorname{Tr}((\mathbf{I} - \mathbf{F})(p\rho - (1 - p)\sigma)) \end{aligned}$$

and, if we note that the eigenvalues of $\mathbf{F}$ must be ≤ 1 , this quantity is maximized by taking $\mathbf{F}$ to be the projector on the positive eigenspace of $p\rho - (1 - p)\sigma$, known as the Helstrom Measurement. Taking $p\rho - (1 - p)\sigma = \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$ the two expressions become

$$1 - p + \sum_{\lambda_i > 0} \lambda_i = p - \sum_{\lambda_i < 0} \lambda_i = \frac{1}{2} + \frac{1}{2} \sum_i |\lambda_i| = \frac{1}{2} + \frac{1}{2} \|p\rho - (1 - p)\sigma\|_{\operatorname{Tr}},$$

known as the Helstrom bound.

Security Scenario: Δ -secrets

Imagine that Alice shares a key with Bob, hiding it from Eve.

Whatever the protocol, at the end we will have a tripartite quantum state containing the key Alice will use, Eve's accrued side information and some other stuff (e.g. Bob's key).

The strongest security requirement is that Eve can't even tell if her memory is entangled with Alice.

So let's set up the following game: Alice agrees to send to Eve her half of ρ_{AE} with probability 50%, on condition that in the other 50% of the cases she sends to Eve $\frac{1}{d_A} \mathbf{I}_A$ and she is allowed to reset Eve's state to any σ_E she wants. Can Eve determine which it was?

We say that ρ_{AE} contains a Δ -secret if Eve's optimal worst-case probability to guess is $\leq \frac{1}{2}(1 + \Delta)$, i.e. if

$$\min_{\sigma_E} \frac{1}{2} \left\| \rho_{AE} - \frac{1}{d_A} \mathbf{I}_A \otimes \sigma_E \right\|_{Tr} \leq \Delta$$

Quantum Leftover Hashing Lemma

For example, if Alice derives her key by locally measuring observable $\mathbf{O}$ upon ρ_A and using a 2-universal hash function h to hash it down to ℓ bits, the Δ -secrecy level is

$$\Delta = \epsilon + \frac{1}{2} \sqrt{2^{\ell - H_{\min}^{\epsilon}(\mathbf{O}|E)_{\rho}}},$$

where for bipartite states ρ_{AE} conditional min-entropy is

$$\begin{aligned} H_{\min}(A | E)_{\rho} &= \max_{\sigma_E} \sup \left\{ \lambda \mid 2^{-\lambda} \mathbf{I}_A \otimes \sigma_E \geq \rho_{AE} \right\} \\ &\leq \sup \left\{ \lambda \mid 2^{-\lambda} \mathbf{I}_A \geq \rho_A \right\} = H_{\min}(A)_{\rho}, \end{aligned}$$

and the conditional smooth min-entropy is

$$\begin{aligned} H_{\min}^{\epsilon}(A | E)_{\rho} &= \max \{ H_{\min}(A | E)_{\tau} \mid P(\tau, \rho) \leq \epsilon \} \\ &\geq H_{\min}(A | E)_{\rho}, \end{aligned}$$

where P is the purified distance.

Universal Hashing in BB84 Variants

In practice, QKD devices use the ϵ for which we have an experimental estimator of $H_{min}^\epsilon(\mathbf{O} | E)$.

For example, the BB84 variant with the following parameters:

- ▶ block length n bits, of which k used to measure mismatch rate
- ▶ $\mathbf{X}, \mathbf{Z}$ sampled so that $\mathbf{X}$ occurs with probability $1/(1 + \sqrt{k/n})$
- ▶ protocol aborted if mismatch rate $\geq Q_{tol}$
- ▶ remaining bits' error correction leaks $\leq leak_{EC}$ bits
- ▶ resulting strings compared using a random 2-universal $\lceil 1/\epsilon_{corr} \rceil$ -bit hash, aborting on mismatch
- ▶ resulting strings hashed down to ℓ -bit hash used for key

has probability ϵ_{corr} to produce mismatching keys and

$$H_{min}^{\epsilon_{corr} + \epsilon_{sec}}(\mathbf{X} | E) \geq n(1 - h(Q_{tol} + \mu)) - leak_{EC} - \log \frac{2}{\epsilon_{corr}},$$

where $\mu = \sqrt{\frac{n+k}{nk} \frac{k+1}{k} \ln \frac{2}{\epsilon_{sec}}}$ i.e. if $\ell \leq RHS + \alpha \log \epsilon_{sec}$, we find

$$\Delta \leq \epsilon_{cor} + (1 + \frac{\alpha}{2})\epsilon_{sec}, \text{ e.g. for } \alpha = 4, \epsilon_{corr} = \epsilon_{sec} = \frac{\epsilon}{2} \Rightarrow \Delta \leq 2\epsilon.$$

Quantum Pinsker Inequality

Notice that the lower the Helstrom bound is, the more similar ρ and σ are. In particular, if we take $p = 1 - p = 1/2$, the Helstrom bound becomes $\frac{1}{2} + \frac{1}{4} \|\rho - \sigma\|_{Tr}$, which we can relate to $D(\rho\|\sigma)$.

So, let's consider all state ρ, σ such that $\|\rho - \sigma\|_{Tr}$ and consider what their minimal Umegaki divergence might be.

To start, consider the channel that performs the Helstrom measurement and records its value in a qubit state.

Letting $\rho - \sigma = \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$ we thus consider

$$\mathcal{E}(\mathbf{M}) = \sum_{\lambda_i > 0} |0\rangle \langle \psi_i| \mathbf{M} |\psi_i\rangle \langle 0| + \sum_{\lambda_i < 0} |1\rangle \langle \psi_i| \mathbf{M} |\psi_i\rangle \langle 1|,$$

and since $\mathcal{E}$ is a channel it lowers the Umegaki divergence.

However, as it turns out the Helstrom bound for discriminating $\mathcal{E}(\rho)$ and $\mathcal{E}(\sigma)$ is $\|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_{Tr} = \|\mathcal{E}(\rho - \sigma)\|_{Tr} = \|\rho - \sigma\|_{Tr}$. It follows that the minimal Umegaki divergence for a given trace distance is attained for diagonal qubit states.

Quantum Pinsker Inequality

For diagonal states ρ, σ the trace distance only depends on their (probabilistic) diagonal vectors $\mathbf{p}, \mathbf{q}$, $\|\rho - \sigma\|_{Tr} = \|\mathbf{p} - \mathbf{q}\|_1$, so we can borrow a result from classical information theory known as the Pinsker inequality which states that

$$\|\mathbf{p} - \mathbf{q}\|_1 \leq \sqrt{2 \ln 2 D_{KL}(\mathbf{p} \parallel \mathbf{q})},$$

and so we find that for any quantum states ρ, σ

$$\begin{aligned} \|\rho - \sigma\|_{Tr} &= \|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_{Tr} = \|\mathbf{p} - \mathbf{q}\|_1 \\ &\leq \sqrt{2 \ln 2 D_{KL}(\mathbf{p} \parallel \mathbf{q})} = \sqrt{2 \ln 2 D(\mathcal{E}(\rho) \parallel \mathcal{E}(\sigma))} \\ &\leq \sqrt{2 \ln 2 D(\rho \parallel \sigma)}, \end{aligned}$$

a relation known as the quantum Pinsker inequality.

We thus successfully related another practically relevant quantity, the maximal probability to distinguish between two states, to an entropy, here the quantum relative entropy or Umegaki divergence.

Entanglement-Assisted Quantum State Discrimination

Mixed states are often a result of tracing out part of an entangled state. Let's say Alice's system has a density matrix either ρ or σ with equal probability, but her system is entangled with Bob's, who is thus able to help Alice. Let's say they share $|\phi_\rho\rangle$ or $|\psi_\sigma\rangle$.

Since there are always $\theta, |\xi\rangle$ such that $|\phi_\rho\rangle = \cos \theta |\psi_\sigma\rangle + \sin \theta |\xi\rangle$,

$$|\phi_\rho\rangle \langle \phi_\rho| - |\psi_\sigma\rangle \langle \psi_\sigma| \sim \begin{bmatrix} \cos^2 \theta - 1 & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$$

with eigenvalues $\pm \sqrt{|\sin^2 \theta (\cos^2 \theta - 1) - \cos^2 \theta \sin^2 \theta|} = \pm \sin \theta$ so the Helstrom bound is $\frac{1}{2}(1 + |\sin \theta|) = \frac{1}{2}(1 + \sqrt{1 - |\langle \phi_\rho | \psi_\sigma \rangle|^2})$.

So, the maximal probability that Alice can discern what was sent depends on the minimal overlap between two purifications, and conversely, the minimal probability that Alice can discern what was sent depends on the maximal overlap between two purifications, a quantity known as the fidelity of the two quantum states.

Uhlmann's Theorem

As it turns out, for any ρ, σ we can compute the fidelity $F(\rho, \sigma)$ as

$$\max \left\{ |\langle \phi_\rho | \psi_\sigma \rangle|^2 \mid \text{Tr}_B |\phi_\rho\rangle \langle \phi_\rho| = \rho, \text{Tr}_B |\psi_\sigma\rangle \langle \psi_\sigma| = \sigma \right\} = \text{Tr}^2 |\sqrt{\rho} \sqrt{\sigma}|.$$

To see why, let's fix some decompositions. Let's take

$$\rho = \sum_i p_i |a_i\rangle \langle a_i|; \quad \sigma = \sum_j q_j |b_j\rangle \langle b_j|;$$

$$|\phi_\rho\rangle = \sum_i \sqrt{p_i} |a_i\rangle \otimes |x_i\rangle; \quad |\psi_\sigma\rangle = \sum_j \sqrt{q_j} |b_j\rangle \otimes |y_j\rangle$$

$$\begin{aligned} \Rightarrow |\langle \phi_\rho | \psi_\sigma \rangle| &= \left| \sum_{i,j} \sqrt{p_i q_j} \langle a_i | b_j \rangle \langle x_i | y_j \rangle \right| \\ &= \left| \sum_{i,j,k,l,m} \langle a_k | \underbrace{\sqrt{p_i} |a_i\rangle \langle a_i|}_{\sqrt{\rho}} \underbrace{\sqrt{q_j} |b_j\rangle \langle b_j|}_{\sqrt{\sigma}} \underbrace{\langle b_l | \langle y_l | \overline{x_m} \rangle \langle a_m |}_{\mathbf{U}} |a_k\rangle \right| \\ &= |\text{Tr}(\sqrt{\rho} \sqrt{\sigma} \mathbf{U})| \leq \text{Tr}(\sqrt{(\sqrt{\rho} \sqrt{\sigma})(\sqrt{\rho} \sqrt{\sigma})^\dagger}) = \text{Tr} |\sqrt{\rho} \sqrt{\sigma}|, \end{aligned}$$

where we used the triangle inequality $|\text{Tr}(\mathbf{A}\mathbf{U})| \leq \text{Tr} |\mathbf{A}|$ is saturated when $\mathbf{A}\mathbf{U}$ is positive semi-definite, e.g. when $|x_i\rangle = |y_i\rangle$.

Entanglement Fidelity

Let's now say that we want to develop a quantum compression algorithm that does not degrade quality of service.

Essentially, we would want Alice to be unable to tell that her states were compressed and decompressed even if she tried.

Alice can get help from any party whom she holds entanglement with, and so what we intuitively want to maximize is the quantum fidelity between ρ and $\mathcal{E}(\rho)$, that is $F(\rho, \mathcal{E}(\rho)) = \text{Tr}^2 \left| \sqrt{\rho} \sqrt{\mathcal{E}(\rho)} \right|$. The problem is that the usual fidelity $F(\rho, \sigma)$ maximizes over both purifications of ρ, σ , whereas our compression $\mathcal{E}$ only acts on Alice's system so Bob does not have free choice over the state of his subsystem, hence we need a more refined type of fidelity.

So, instead, we are going to consider the entanglement fidelity,

$$\begin{aligned} F(\rho, \mathcal{E}) &= F(\mathbf{I}_{Bob} \otimes \mathcal{E}_{Alice}(|\psi_\rho\rangle \langle \psi_\rho|), |\psi_\rho\rangle \langle \psi_\rho|) \\ &= \langle \psi_\rho | \mathbf{I}_{Bob} \otimes \mathcal{E}_{Alice}(|\psi_\rho\rangle \langle \psi_\rho|) | \psi_\rho \rangle, \end{aligned}$$

where any chosen purification $|\psi_\rho\rangle$ produces the same result.

Quantum Fano Inequality

When, as before, σ is pure i.e. $\sigma = |\phi\rangle\langle\phi|$ we find

$$\begin{aligned} F(\rho, \sigma) &= \text{Tr} |\sqrt{\rho}\sqrt{\sigma}| = \text{Tr} \sqrt{(\sqrt{\rho}\sqrt{\sigma})^\dagger (\sqrt{\rho}\sqrt{\sigma})} = \text{Tr} \sqrt{\sigma}^\dagger \rho \sqrt{\sigma} \\ &= \text{Tr} |\phi\rangle\langle\phi| \rho |\phi\rangle\langle\phi| = \langle\phi| \rho |\phi\rangle, \end{aligned}$$

numerically equal to the probability that ρ would collapse into $|\phi\rangle$. If Alice and Bob try to evaluate the entanglement fidelity $F(\rho, \mathcal{E})$, they thus can simply measure upon $\rho' = \mathbf{I} \otimes \mathcal{E}(|\psi_\rho\rangle\langle\psi_\rho|)$ a POVM with operators $\{|\psi_\rho\rangle\langle\psi_\rho|, \mathbf{I} - |\psi_\rho\rangle\langle\psi_\rho|\}$ and they may for example record the outcome in a single-qubit (diagonal) state ρ'' .

Looking at the entropy exchange when ρ goes through $\mathcal{E}$ we get

$$\begin{aligned} H(\rho, \mathcal{E}) &= H(\rho') - H(|\psi_\rho\rangle\langle\psi_\rho|) \\ &= \log d^2 - D(\rho' \mid \mathbf{I}_d/d^2) - 0 \\ &\leq \log d^2 - D(\rho'' \mid \frac{1}{d^2} |0\rangle\langle 0| + \frac{d^2 - 1}{d^2} |1\rangle\langle 1|) \\ &= h(F(\rho, \mathcal{E})) + (1 - F(\rho, \mathcal{E})) \log(d^2 - 1), \end{aligned}$$

known as the Quantum Fano Inequality.

Schumacher's Source Coding Theorem

With our new tricks, we can now look at quantum compression. As with Shannon's source coding theorem, what we are going to be compressing is not a single state but a (possibly infinite) stream of i.i.d. samples of a random variable with matrix $\rho \in \mathcal{B}(\mathcal{H})$.

But, to define encoder and decoder channels we need to know how many such samples n we are processing at a time, and how many qubits m we may output. We say that a pair of channels

$$\mathcal{E} : \mathcal{B}(\mathcal{H}^{\otimes n}) \rightarrow \mathcal{B}(\mathbb{C}^{2^{\otimes m}}); \mathcal{D} : \mathcal{B}(\mathbb{C}^{2^{\otimes m}}) \rightarrow \mathcal{B}(\mathcal{H}^{\otimes n})$$

is a (n, m, π) compression scheme for ρ if $F(\rho^{\otimes n}, \mathcal{D} \circ \mathcal{E}) \geq 1 - \pi$.

Let's now say we are given samples in bulk ($n \rightarrow \infty$), and we want to know how many qubits R we need per sample on average, i.e. the minimal R such that as $n \rightarrow \infty$ we can always find a $(n, nR + o(n), \pi_n)$ compression scheme with $\pi_n \rightarrow 0$.

Schumacher's Source Coding Theorem states that $\min R = H(\rho)$. Note that lower R would let us violate the Holevo bound.

Schumacher's Source Coding Theorem

As with Shannon's source coding theorem, to show that we can achieve $R = H(\rho)$ we want to look at some ϵ -typical set.

If $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$, then

$$\rho^{\otimes n} = \sum_{i_1, \dots, i_n} p_{i_1} \dots p_{i_n} |\psi_{i_1}\rangle \langle \psi_{i_1}| \otimes \dots \otimes |\psi_{i_n}\rangle \langle \psi_{i_n}|$$

and so we denote by $\mathcal{I}$ the set of ϵ -typical indices $(i_1, \dots, i_n)$.

Consider now the projector $\mathbf{P} = \sum_{\mathbf{i} \in \mathcal{I}} |\psi_{i_1}\rangle \langle \psi_{i_1}| \otimes \dots \otimes |\psi_{i_n}\rangle \langle \psi_{i_n}|$, and notice that if we measure $\rho^{\otimes n}$ in its eigenbasis, the probability to obtain a result in $\mathcal{I}$ is $\text{Tr}(\mathbf{P}\rho) \geq 1 - \pi$.

Yet, from the proof of Shannon's theorem we know that if we were to simply send such an index it would take $n(H(\rho) + C\epsilon)$ bits, so to have them as base states we would need $n(H(\rho) + C\epsilon)$ qubits. Thus, we choose to encode our state using an isometry that maps the ϵ -typical eigenvectors to base states of our target system.

As it turns out, this is enough to achieve $F(\rho^n, \mathcal{D} \circ \mathcal{E}) \geq 1 - \pi$.

Follow-up Reading

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- ▶ N.Sharma – Extensions of the quantum Fano inequality.
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- ▶ A. Müller-Hermes – Quantum information theory (PDF),
Schumacher's compression theorem and quantum entropy